MSQF 533 FINANCIAL RISK MANAGEMENT

UNIT III: CREDIT RISK MANAGEMENT AND DERIVATIVES

1. Introduction to Credit Risk

1.1 Meaning of credit risk

Credit risk is the possibility that a borrower, bond issuer, counterparty or other party will fail to meet a financial obligation as agreed. The obligation may be the repayment of a loan, payment of interest, repayment of a bond, or settlement of a derivative contract.

In simple words, credit risk means: "Will the other party pay what it owes, in full and on time?"

Example: A bank lends ₹10,00,000 to a small business. If the business closes and cannot repay the loan, the bank faces credit risk. Even if the borrower pays eventually, late payment can create cash-flow and funding problems.

1.2 Why credit risk matters

- Banks earn interest by lending, but a portion of borrowers may default.
- A default can reduce profit, erode capital and threaten a bank's ability to lend.
- Credit losses can spread through financial markets when institutions are connected.
- Investors face credit risk when they buy corporate bonds, commercial paper or other debt securities.
- Derivatives can create counterparty credit risk even when no ordinary loan exists.

1.3 Main causes of credit risk

Cause	Explanation	Illustration
Borrower-specific	Poor management, fraud, excessive debt or loss of a major customer.	A company loses its largest buyer and cannot service its loan.
Economic	Recession, inflation, high interest rates or falling demand.	A rise in interest rates increases EMI burden for floating-rate borrowers.
Industry	Changes in technology, regulation or commodity prices.	A sudden fall in steel prices weakens a highly leveraged steel producer.
Collateral-related	Collateral value falls or cannot be sold quickly.	Property worth ₹50 lakh at loan origination may later sell for ₹38 lakh.

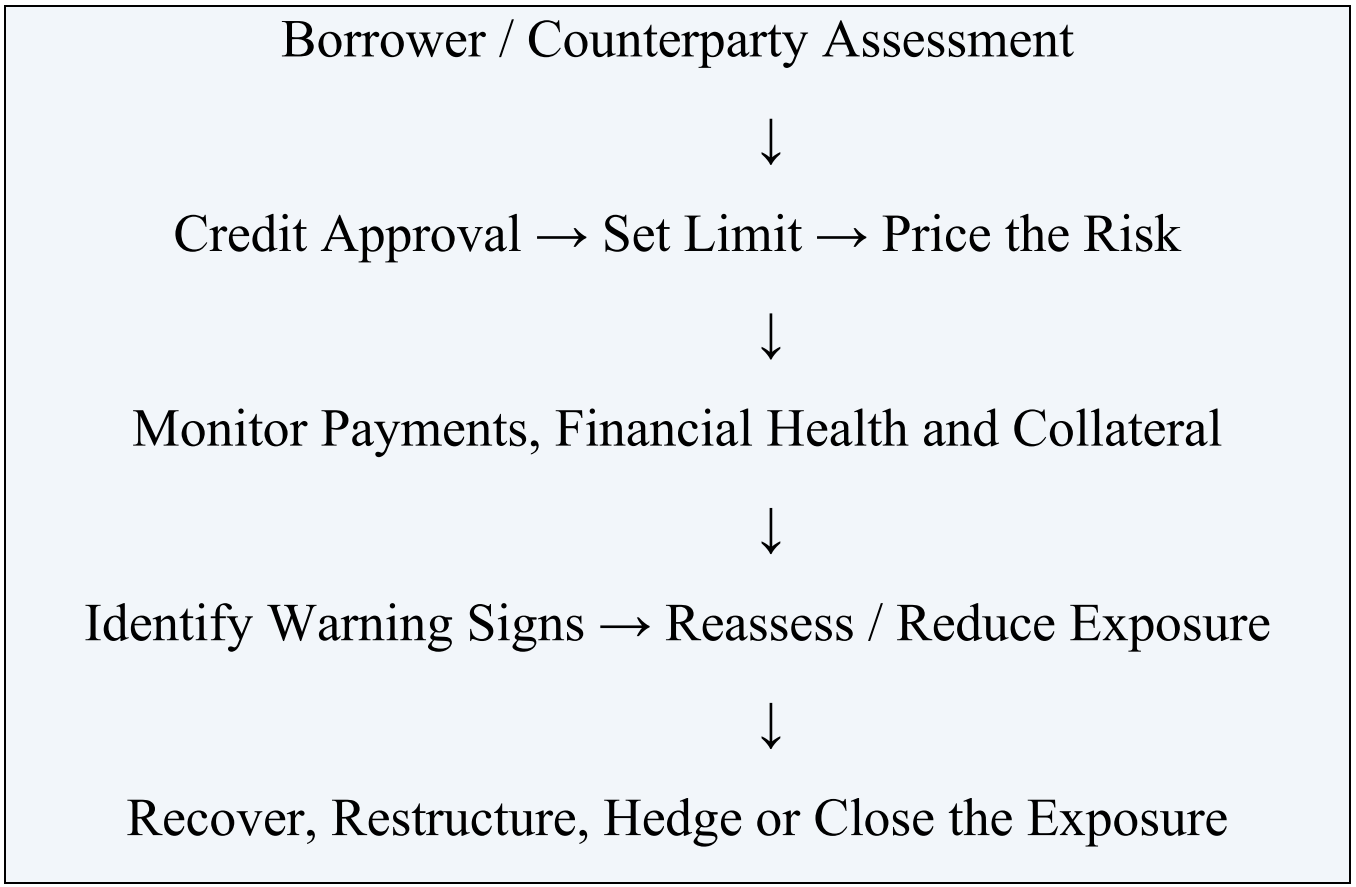
Concentration	Too much exposure to one borrower, group, sector or region.	A bank lends heavily to real estate and suffers when property prices fall.
Counterparty	A party to a market transaction fails to pay.	A derivative counterparty defaults when the contract has positive value to the bank.

1.4 Types of credit risk

Type	Meaning	Example
Default risk	The obligor fails to pay principal or interest.	A company misses a bond coupon and enters default.
Credit migration risk	The borrower's credit quality deteriorates, even without default.	A bond is downgraded from A to BBB and its market price falls.
Counterparty credit risk	A trading counterparty fails before final settlement of a transaction.	A bank's derivative counterparty defaults while the derivative is an asset to the bank.
Concentration risk	Losses are amplified by exposures linked to one name or risk factor.	Several borrowers depend on the same agricultural crop.

Country/sovereign risk	Political, economic or transfer restrictions prevent payment.	A government imposes capital controls that delay cross-border repayment.
Settlement risk	One party pays but does not receive the expected asset or payment.	A bank delivers currency but does not receive the other currency.

1.5 Credit risk management process



The process is continuous. A borrower who was safe last year may become risky after a business downturn, a change in ownership or a sharp increase in debt.

1.6 Credit risk versus market risk

Feature	Credit risk	Market risk
Core question	Will the other party pay?	Will market prices move adversely?
Main drivers	Default, credit quality, counterparty strength	Interest rates, share prices, FX rates, commodity prices, volatility
Typical loss	Loan principal or interest not recovered	Loss in value of a trading position
Example	Corporate borrower defaults on a loan	Bond price falls when market yields rise
Relationship	A downgrade can also reduce bond prices	Market movements can increase counterparty exposure

2. Measuring Credit Risk

2.1 The three building blocks: PD, LGD and EAD

A common starting point for measuring credit risk is Expected Loss (EL). It combines the chance of default, the size of exposure at default, and the proportion of that exposure likely to be lost.

Term	Full form	Simple meaning
PD	Probability of Default	Chance that the borrower defaults over a specified period.
LGD	Loss Given Default	Percentage of the exposure that is not recovered after default.

EAD	Exposure at Default	Amount the lender is exposed to when default occurs.

Expected Loss (EL) = Probability of Default (PD) × Loss Given Default (LGD) × Exposure at Default (EAD)

Important: PD is expressed as a decimal in calculations (5% = 0.05), and LGD is also entered as a decimal. EAD is an amount of money.

2.2 Worked example: Expected loss on a loan

A bank has lent ₹20,00,000 to a company. The estimated one-year probability of default is 3%. If default occurs, the bank expects to lose 40% of the exposure after recoveries. Calculate expected loss.

Step 1: PD = 3% = 0.03. Step 2: LGD = 40% = 0.40. Step 3: EAD = ₹20,00,000.

EL = 0.03 × 0.40 × ₹20,00,000 = ₹24,000.

Interpretation: The estimated average credit loss for this exposure over the specified horizon is ₹24,000. This is a statistical expectation, not a prediction that exactly ₹24,000 will be lost.

2.3 Worked example: Recovery rate and LGD

Suppose a lender has an exposure of ₹5,00,000. After default, it recovers ₹3,25,000 through payments, collateral sale and other recovery actions.

Recovery rate = Recovery amount ÷ EAD = ₹3,25,000 ÷ ₹5,00,000 = 65%.

LGD rate = 1 − Recovery rate = 1 − 0.65 = 35%.

Loss amount = ₹5,00,000 × 35% = ₹1,75,000.

2.4 Estimating Probability of Default

PD can be estimated from historical default rates, borrower financial ratios, credit ratings, credit scores, market indicators and statistical models. The estimate must match the relevant borrower type, product and time horizon.

Approach	How it works	Limitation
Historical default rate	Uses observed defaults among similar borrowers.	Past conditions may not represent the future.
Credit scoring	Combines factors such as repayment history, debt, income and leverage.	Scores depend on data quality and model design.
Rating-based	Uses default experience associated with a credit-rating grade.	Ratings may change slowly and are not guarantees.

Market-implied	Uses bond spreads or CDS spreads to infer market perceptions of credit risk.	Spreads also reflect liquidity, risk appetite and other premia.
Statistical / machine learning	Estimates default probability from borrower and macroeconomic variables.	Can overfit, be hard to explain or perform poorly during stress.

2.5 Credit exposure

Credit exposure is the amount a lender or counterparty could lose if the other party fails to meet its obligation. The definition depends on the product.

Product	How exposure is viewed
Term loan	Outstanding principal plus applicable accrued amounts, subject to the measurement method.
Credit card / overdraft	Current drawn balance plus an estimate of future drawings before default.
Guarantee / letter of credit	Potential amount the bank may have to pay if the guarantee is called.
Derivative	Current positive replacement value plus a measure of possible future exposure.
Bond investment	Value of the debt claim exposed to issuer credit deterioration or default.

2.6 Current exposure and potential future exposure

For a derivative, current exposure is often related to the positive replacement value of the contract after taking eligible netting and collateral into account. Potential future exposure (PFE) estimates how much the exposure could increase over a future period due to market movements.

Illustration: A derivative has a positive replacement value of ₹4 lakh today. Market movements could raise its value to ₹7 lakh before the contract ends. The current exposure is approximately ₹4 lakh before relevant adjustments, while the possible future exposure is represented by an appropriate model or regulatory add-on—not automatically the full ₹7 lakh.

2.7 Credit conversion factor (CCF)

A Credit Conversion Factor estimates how much of an undrawn or contingent commitment may become an exposure. The applicable CCF depends on the type of facility and the relevant regulatory rules.

Illustration: A borrower has an undrawn credit line of ₹10 lakh. If an illustrative CCF of 40% is assumed, the converted amount is ₹10 lakh × 40% = ₹4 lakh. This is a teaching example only; actual regulatory CCFs must be checked against the applicable rules.

2.8 Expected loss versus unexpected loss

Concept	Meaning	Management response
Expected loss (EL)	Average loss anticipated over a specified period.	Typically reflected in pricing, provisions and business planning, subject to accounting and regulatory rules.
Unexpected loss (UL)	Loss above the expected level, arising from variability and adverse outcomes.	Capital and risk limits help absorb losses beyond the expected level.

A bank may expect small losses across many loans, but a recession can cause many borrowers to default together. The resulting loss may be much larger than the average expected loss.

3. Credit Derivatives

3.1 Meaning and purpose

A credit derivative is a financial contract through which credit risk is transferred from one party to another. The underlying loan or bond may remain with the original lender, while the credit risk is partly or fully transferred according to the contract.

Think of it as a contract that separates the ownership of a debt asset from some of the risk that the borrower will fail to pay.

3.2 Why institutions use credit derivatives

- Hedge credit risk without necessarily selling the underlying loan or bond.
- Manage concentration in a borrower, sector or country.
- Take a view on changes in credit quality or credit spreads.
- Improve portfolio flexibility, subject to counterparty, legal and basis risks.
- Transfer specified credit risk to investors willing to bear it.

3.3 Main types of credit derivatives

Instrument	How it works	Example / use
Credit Default Swap (CDS)	Protection buyer pays a premium; protection seller pays if a defined credit event occurs, subject to contract terms.	A bank buys protection on a corporate bond issuer.
Total Return Swap (TRS)	One party transfers the total economic return of a reference asset; the other pays a financing rate or other agreed return.	An investor obtains bond exposure without buying the bond outright.
Credit Spread Option	Payoff depends on a credit spread reaching or moving beyond a specified level.	An investor hedges against a widening corporate bond spread.

Credit-Linked Note (CLN)	A funded note whose repayment or return is linked to credit events on a reference entity or portfolio.	Investors receive a coupon but may lose principal after a defined credit event.
Basket / portfolio credit derivative	Credit protection or payoff references multiple entities.	A portfolio investor manages or takes exposure to a group of borrowers.

3.4 Credit derivatives: advantages and risks

Potential benefits	Risks and limitations
Transfer specified credit risk	Protection seller may itself default.
Manage a concentrated exposure	Basis risk: hedge may not match the actual asset.

Keep the customer relationship or asset	Legal documentation and credit-event definitions can be complex.
Adjust exposure more flexibly than by selling a loan	Market liquidity may disappear in stress.
Create credit-risk investment opportunities	Leverage, valuation uncertainty and interconnected exposures can amplify losses.

4. Credit Default Swap (CDS)

4.1 Meaning

A Credit Default Swap is a contract in which the protection buyer pays periodic premiums to the protection seller. In exchange, the seller agrees to compensate the buyer if a contractually defined credit event occurs for a reference entity or obligation.

A CDS resembles insurance in its broad purpose, but it is a financial derivative with specific contractual definitions, valuation rules, counterparty risk and legal requirements. It should not be treated as identical to an ordinary insurance policy.

4.2 Parties to a CDS

Party / term	Meaning
Protection buyer	Pays the CDS premium and receives protection against defined credit-event losses.

Protection seller	Receives the premium and assumes the contractual credit-event payment obligation.
Reference entity	The borrower or issuer whose credit risk is referenced.
Reference obligation	A specified bond, loan or other debt obligation used in the contract.
Credit event	A contractually defined event, potentially including failure to pay, bankruptcy or restructuring, depending on the contract.
Notional amount	Reference amount used to calculate premiums and settlement payments; it is not necessarily the amount exchanged at the start.

4.3 CDS example: protection payment

A bank owns corporate bonds with a face value of ₹1 crore. It buys CDS protection on the same reference entity for a notional amount of ₹1 crore. Following a covered credit event, the relevant settlement recovery value is determined to be 35% of par under the applicable settlement mechanism.

Simplified cash-settlement illustration: Payment ≈ Notional × (1 − Recovery rate) = ₹1,00,00,000 × (1 − 0.35) = ₹65,00,000.

The ₹65 lakh is an illustrative protection payment before considering contract-specific terms, accrued premium, deliverability, auction rules, settlement costs, counterparty failure and any mismatch between the bond and CDS.

4.4 CDS as a hedge

Suppose an investor owns a corporate bond worth ₹50 lakh. The investor is worried that the issuer's credit quality may deteriorate. Buying CDS protection may reduce the credit-event loss on a matching notional amount.

Without CDS protection	With CDS protection
Bond price may fall after a downgrade or spread widening.	A CDS position may gain value as the credit spread widens.
Default can cause a large loss, depending on recovery.	A covered credit event can trigger a contractual payment.
Investor bears issuer credit risk directly.	Investor exchanges some issuer credit risk for CDS counterparty, basis, liquidity and contract risks.

A CDS hedge is not perfect automatically. The bond and CDS may differ in maturity, seniority, currency, restructuring terms, deliverability or spread behaviour. These differences create basis risk.

4.5 CDS advantages and disadvantages

Advantages	Disadvantages
Transfers defined credit-event risk	Does not eliminate all risks of the underlying bond.
Can hedge without selling the asset	Protection seller may default.
Can help manage concentration	May be costly if spreads are wide.
Can provide a market signal of credit concerns	Spreads can be affected by liquidity and market stress.
Allows tailored reference entities and maturities	Contract terms and settlement mechanics can be complex.

5. Credit Risk Hedging and Credit Derivatives

5.1 What is hedging?

Hedging means taking a position designed to offset all or part of an existing risk. A hedge reduces a targeted risk but may introduce other risks or costs.

5.2 Common credit hedging methods

Method	Description	Main limitation
Collateral	Obtain assets that can be enforced or sold after default.	Collateral value may fall; enforcement takes time.
Guarantee	A third party promises to pay if the borrower fails to do so.	The guarantor may also become unable to pay.
Credit insurance / guarantee schemes	Contractual compensation for specified losses.	Coverage exclusions and claim conditions apply.

Credit Default Swap	Transfer defined credit-event risk to a protection seller.	Counterparty, basis, liquidity and legal risks remain.
Syndication / loan sale	Share or transfer a loan exposure to other lenders or investors.	May be costly or difficult in illiquid markets.
Diversification and limits	Avoid excessive dependence on a borrower or sector.	System-wide shocks can affect many exposures together.

5.3 Important risks after hedging

- Residual risk: only part of the exposure may be hedged.
- Basis risk: the hedge value and underlying exposure do not move together.
- Counterparty risk: the hedge provider fails.
- Liquidity risk: the hedge cannot be unwound at a fair price.
- Legal risk: the contract or collateral arrangement may not be enforceable.
- Wrong-way risk: the hedge provider becomes weaker when the underlying exposure becomes riskier.

6. Credit Risk Models

6.1 Why use models?

Credit risk models turn information about borrowers, exposures, defaults and recoveries into estimates that support lending decisions, pricing, portfolio limits, provisions and capital planning. A model helps structure decisions; it does not remove uncertainty or replace sound judgment.

6.2 Credit scoring models

Credit scoring assigns a score or risk grade using borrower information. Typical inputs include repayment history, debt-to-income ratio, leverage, liquidity, profitability, business age and credit history.

Example: A lender may assign points for timely repayment and stable income, but deduct points for high debt, repeated late payments or weak cash flow. A lower score may lead to rejection, a smaller loan, more collateral or a higher interest rate.

Strengths: relatively fast, consistent and scalable. Limitations: data quality, model bias, changing economic conditions and limited ability to capture unusual events.

6.3 Structural models

Structural models connect default risk to the economic value of a firm's assets and its liabilities. In the classic Merton-style intuition, equity can be viewed like a call option on firm assets, while default becomes more likely when asset value is insufficient to meet debt obligations at maturity.

Illustration: A firm has assets valued at ₹100 crore and debt due of ₹60 crore. If asset value falls sharply to ₹45 crore and the firm cannot refinance or raise funds, the risk of default rises. Real-world default depends on timing, liquidity, debt structure and legal conditions—not asset value alone.

6.4 Reduced-form models

Reduced-form models estimate default as a probabilistic event using factors such as credit spreads, rating, macroeconomic conditions and a default intensity (hazard rate). They do not require default to be explained directly by a detailed model of the firm's asset value.

Example: A model may estimate that a company has a higher default intensity after its CDS spread widens, its rating is downgraded and industry conditions weaken.

6.5 Portfolio credit risk models

Portfolio models estimate the distribution of losses across many exposures. They consider borrower default probabilities, exposure amounts, loss severities and the degree to which defaults are correlated.

Why correlation matters: if 20 borrowers are independent, their defaults may not occur together very often. If all 20 rely on the same industry, a single industry shock may affect many at once, producing a much larger portfolio loss.

6.6 Migration / rating transition models

A rating transition matrix shows the probability that an issuer moves from one rating category to another over a stated horizon, including the possibility of default. The rows typically represent the starting rating; columns show the rating at the end of the period.

Starting rating	Remain same	Downgrade	Upgrade	Default
A	High probability	Some probability	Some probability	Low probability
BBB	Moderate probability	Meaningful probability	Some probability	Higher than A
B	Lower probability	Meaningful probability	Possible	Higher than investment-grade categories

This table is conceptual rather than empirical. Actual transition probabilities must come from a stated rating agency, dataset, time period and methodology.

6.7 Stress testing and scenario analysis

Stress testing asks what happens if severe but plausible conditions occur. Examples include a recession, a 300-basis-point increase in interest rates, a fall in property prices, a currency shock or a simultaneous deterioration in several major borrowers.

Illustration: A bank's expected loss is ₹5 crore in the base case. Under a severe recession scenario, PDs rise, collateral values fall and LGDs increase. The stressed loss might be much higher. The result helps management assess whether limits, provisions, capital and contingency plans are adequate.

6.8 Model risk and validation

- Check whether input data are accurate, complete and representative.
- Compare model predictions with actual defaults and losses.
- Test performance across borrower groups and economic periods.
- Document assumptions and limitations.
- Monitor model drift and update models when conditions change.

- Use independent review and human oversight for material decisions.

7. Basel Accord and Credit Risk Regulation

7.1 What is the Basel framework?

The Basel framework consists of international banking standards developed by the Basel Committee on Banking Supervision. It aims to strengthen banks' capital, risk management, supervisory review and disclosure. Local regulators implement the applicable requirements through their own rules.

7.2 Basel I, Basel II and Basel III

Framework	Main emphasis	Simplified explanation
Basel I (1988)	Minimum capital against credit risk using broad risk weights.	Introduced a common international capital framework; the original standard set an 8%

		minimum total capital ratio against risk-weighted assets.
Basel II (2004; later revisions)	Three pillars: minimum capital, supervisory review, market discipline.	Made credit-risk measurement more risk-sensitive and developed operational-risk capital requirements.
Basel III (post-global financial crisis)	Higher-quality capital, buffers, leverage ratio, liquidity standards and stronger risk coverage.	Improves resilience and addresses weaknesses exposed by the global financial crisis; implementation details vary by jurisdiction and date.

7.3 The three pillars of Basel II

Pillar	Meaning	Illustration
Pillar 1: Minimum capital requirements	Rules for credit risk, market risk and operational risk.	A bank calculates regulatory capital requirements using an allowed approach.
Pillar 2: Supervisory review	Supervisors assess whether the bank's capital and risk management are adequate for its full risk profile.	A supervisor may require improvements if the bank has concentrated exposures or weak stress testing.
Pillar 3: Market discipline	Disclosure helps investors and other market participants understand a bank's risks and capital.	The bank publishes information about capital, risk exposures and risk-management practices.

7.4 Risk-weighted assets (RWA)

Risk-weighted assets are exposures adjusted by regulatory risk weights under the applicable framework. Different exposures can receive different weights based on factors such as counterparty type, credit quality, collateral and the regulatory approach.

Simplified illustration: If a bank has ₹100 crore of an exposure with an illustrative 100% risk weight, its RWA contribution is ₹100 crore. If a different eligible exposure has an illustrative 50% risk weight, ₹100 crore of that exposure contributes ₹50 crore of RWA. These weights are examples for learning; real weights depend on the applicable rules and exposure characteristics.

7.5 Capital adequacy ratio

Capital Adequacy Ratio (CAR)

= Eligible regulatory capital ÷ Risk-weighted assets × 100%

Illustration: Eligible capital is ₹12 crore and RWA is ₹100 crore. CAR = ₹12 crore ÷ ₹100 crore × 100% = 12%. Whether this is sufficient depends on the applicable minimums, buffers, supervisory requirements and bank-specific requirements.

7.6 Credit risk capital approaches

Approach	General idea
Standardised Approach	Uses prescribed risk weights and external ratings or other criteria where permitted by the rules.
Internal Ratings-Based (IRB) approaches	Eligible banks use approved internal estimates and regulatory formulas, subject to strict requirements and supervisory approval.
Credit risk mitigation	Recognises eligible collateral, guarantees, netting or other techniques under prescribed conditions.

The ability to use a particular approach depends on regulatory permission, jurisdiction and current implementation. Internal estimates cannot simply be used without meeting approval and validation requirements.

7.7 Credit risk versus market risk capital

Credit risk capital	Market risk capital
Addresses losses from borrower / counterparty credit deterioration and default.	Addresses losses from adverse market-price movements in covered positions.
Often uses PD, LGD, EAD, ratings, risk weights or approved models.	Uses risk sensitivities, risk factors, prescribed calculations or approved market-risk models.
A loan default is a typical event.	An adverse movement in bond yields, FX rates or equity prices is a typical event.
Credit spread changes can also reduce the market value of a traded bond.	A bond may experience both spread-related market losses and issuer-default losses.

8. Settlement Risk

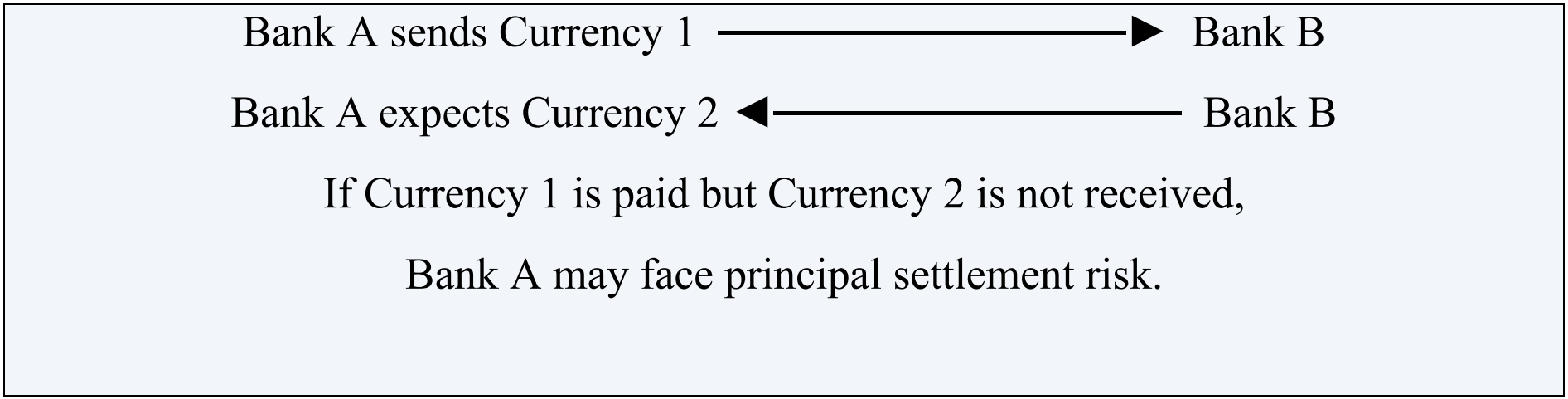
8.1 Meaning

Settlement risk is the risk that a party to a financial transaction delivers money or securities but does not receive the corresponding payment or asset as expected. It can arise in foreign exchange, securities, derivatives and payment systems.

In simple words, one party has completed its side of the deal, but the other party does not complete its side.

8.2 Foreign-exchange settlement example

Bank A agrees to pay US$10 million to Bank B and receive the equivalent amount in another currency. If Bank A sends the dollars but Bank B fails before delivering the other currency, Bank A may lose the full amount sent, depending on the settlement arrangements. This is often described as principal settlement risk.



8.3 Types and related concepts

Risk / concept	Meaning	Example
Principal risk	One party pays the full principal but does not receive the counter-value.	One currency is delivered and the other is not.
Replacement-cost risk	A counterparty fails before settlement; replacing the trade costs more because market prices have moved.	A forward contract is replaced at a less favourable exchange rate.
Liquidity risk	A delayed or failed payment creates a short-term cash shortage.	A bank must obtain emergency funding because expected funds do not arrive.
Operational risk	Processing, system, messaging or human errors disrupt settlement.	Incorrect settlement instructions send funds to the wrong account.
Legal risk	Unclear or unenforceable rules create uncertainty about finality or recovery.	A dispute delays the transfer of securities or funds.

8.4 Causes of settlement risk

- Different time zones and payment-system operating hours.
- Failure of a counterparty or settlement bank.
- Incorrect instructions, fraud, cyber incidents or operational outages.
- Lack of simultaneous payment-versus-payment or delivery-versus-payment mechanisms.
- Insufficient liquidity or collateral.
- Legal uncertainty over settlement finality and close-out netting.

8.5 Methods of reducing settlement risk

Method	How it helps
Payment-versus-Payment (PvP)	In eligible FX arrangements, final transfer of one currency occurs if and only if the other currency transfer occurs.
Delivery-versus-Payment (DvP)	In securities settlement, delivery of securities is linked to payment so that one side does not complete alone.
Central counterparty (CCP) clearing	A CCP interposes itself between counterparties for eligible cleared transactions and manages risk through margin, default resources and procedures.

Netting	Offsets eligible obligations so fewer or smaller payments are made, subject to enforceability and rules.
Collateral and margin	Provides a buffer against losses and helps manage exposure, though it does not eliminate all settlement risk.
Counterparty limits and monitoring	Restricts exposure to counterparties and monitors their financial strength.
Reconciliation and operational controls	Checks that instructions, amounts, accounts and confirmations match.